

ALEX ARIZA HERRERA

AI Engineer | LLM & RAG Systems | Full-Stack Development

Based in Colombia | Remote-First & Onsite Opportunities | arizah2020@gmail.com

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Professional Summary

AI Engineer focused on LLM and RAG systems for regulated domains—healthcare, public sector, and confidential document knowledge—where a wrong answer has real cost. Delivered a compliance-first dental EHR on AWS with HL7 FHIR and MinSalud IHCE integration; a local RAG engine with hybrid retrieval and page-level citations; and a multi-agent SGR/MGA formulation pipeline with versioned normative canon and mandatory human review. Full-stack from ingestion to deployment.

Experience

Full-Stack Engineer — ValexSalud

Feb 2026 – Jul 2026

- Built a compliance-first dental EHR for Colombian clinics: append-only clinical folios, certified digital signatures (PKCS#7/CAdES), RBAC, and immutable audit trails aligned with Law 2015/2020 and Resolution 1888/2025.
- Implemented HL7 FHIR RDA submission to IHCE (MinSalud) with transactional outbox, VIDA tracking, and in-clinic patient record delivery with witness audit.
- End-to-end AWS deployment (EC2, RDS PostgreSQL, S3, ECR, GitHub Actions); OpenAPI as contract between FastAPI and Next.js.

AI Engineer — Academia Naval de Estudios Estratégicos

Jun 2025 – Jan 2026

- Built a local-first RAG knowledge copilot for confidential contracts, policies, and manuals: cited Q&A without sending document content to cloud APIs by default.
- Designed hybrid semantic + lexical retrieval with RRF fusion and heading-aware ranking for Spanish corpora; SHA-256 deduplication and explicit no-evidence responses when the corpus lacks support.
- Delivered FastAPI + React stack with PDF ingestion (Docling), ChromaDB + SQLite FTS5, on-machine inference via llama.cpp, golden-set eval harness, and Prometheus metrics on ingestion, retrieval, rerank, and generation.

Independent Software Engineer & Consultant

Jan 2024 – Present

- **SGR/MGA consultancy:** multi-agent formulation pipeline for territorial pre-investment—15-step deterministic flow, versioned normative canon, evidence-first artifacts, and mandatory human review before methodological closure.
- Built an enterprise RAG copilot over internal PDFs (FastAPI, pgvector, SSE streaming, page-level citations) with JWT auth, per-user document ownership, and a no-answer policy when evidence is missing.
- **Castol Mantenimientos y Servicios S.A.S.:** development and support of internal tools for operational management, information processing, and task automation; systems maintenance, access management, and resolution of technical incidents.
- **Buddi Limited:** development and maintenance of backend components and internal applications, including APIs, data persistence, authentication, and service integrations.
- Full cycle for own products and clients—architecture, APIs, frontend, CI/CD, observability—with public case studies at alexariza.dev.

Education

Universidad de los Andes — Electronic Engineering Degree

The University of Texas at Austin — Certificate in Full Stack Software Development:
Building Scalable Cloud Applications

Skills

Languages: Python, TypeScript, SQL

AI & LLM: RAG, Hybrid Retrieval, Vector Search, Embeddings, Agent Orchestration, Prompt Engineering, Local LLM Inference (llama.cpp), Eval Harnesses

Backend: FastAPI, PostgreSQL, pgvector, SQLAlchemy, Redis, Celery, REST APIs, JWT Auth, RBAC, OpenAPI

Frontend: React, Next.js, TypeScript, Tailwind CSS, TanStack Query

Infrastructure & Quality: Docker, AWS (EC2, RDS, S3, ECR), GitHub Actions, CI/CD, Prometheus, pytest